

# Business Summary Report: Predictive Insights for Collections Strategy

## 1. Summary of Predictive Insights

- Customers with high credit utilization, multiple missed payments, and low credit scores represent the highest-risk delinquency segment.
- Customers with high loan balances and lower income levels show increased repayment risk due to higher debt burden.
- Historical repayment behaviour is a strong predictor of future delinquency.

| Key Insight                                                | Customer Segment                                             | Influencing Variables                             | Potential Impact               |
|------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------|--------------------------------|
| High delinquency risk among financially stressed borrowers | Customers with high credit utilization and low credit scores | Credit Utilization, Credit Score, Missed Payments | Proactive collections outreach |
| Repayment history predicts future behaviour                | Customers with previous late payments                        | Missed Payments, Payment History                  | Early risk identification      |
| Debt burden affects repayment capacity                     | Customers with high loan balances and lower income           | Loan Balance, Income, Debt-to-Income Ratio        | Targeted payment assistance    |

## 2. Recommendation Framework

Restated Insight: High credit utilization significantly increases delinquency risk.

Proposed Recommendation: Implement an AI-driven Early Warning Program for high-utilization customers.

Specific: Identify customers above a defined utilization threshold and provide proactive outreach.

Measurable: Reduce delinquency within this segment by 15%.

Actionable: Generate weekly high-risk customer lists and targeted interventions.

Relevant: Supports collections performance and reduces financial losses.

Time-bound: Launch next quarter and evaluate results after six months.

Justification and Business Rationale: Early intervention can improve repayment outcomes and reduce collections costs.

## 3. Ethical and Responsible AI Considerations

- Fairness Risk 1: Socioeconomic variables may indirectly disadvantage certain groups. Mitigation: monitor model performance and false positive rates across customer segments.
- Fairness Risk 2: Historical data may contain existing biases. Mitigation: retrain regularly

and include human oversight.

- Explainability: Logistic Regression allows clear explanation of why a customer is classified as high risk.
- Responsible AI: Promote transparency, accountability, privacy protection, and human review of recommendations.